

Akshat Sureshbhai Desai

714-519-7380 | akshat.desai.754@gmail.com | linkedin.com/in/akshat-desai | github.com/Akshat1661 | Portfolio

EDUCATION

California State University, Fullerton <i>Master of Science in Computer Science GPA: 3.60/4.00</i>	Fullerton, CA Aug 2024 – May 2026
Charotar University of Science and Technology (CHARUSAT) <i>Bachelor of Technology in Computer Science GPA: 3.58/4.00</i>	Gujarat, India Nov 2020 – May 2024

EXPERIENCE

Lead AI Developer (MLOps & Edge Deployment) <i>Lightwall Art Installation</i>	Oct 2025 – Jan 2026 Fullerton, CA
- Deployed a real-time multimodal inference pipeline, integrating Llama 3.2 with hardware sensors to dynamically control 12 synchronized actuators. - Engineered an asynchronous C++ bridge to translate neural outputs into control packets, optimizing streaming latency via predictive buffering. - Architected a sub-500ms input error-detection and filtering system to eliminate false triggers, achieving reliable, sub-second end-to-end inference latency.	
Graduate Research Assistant (HPC & Pipeline Automation) <i>Cal State Fullerton (Medical Imaging Lab)</i>	Nov 2024 – Present Fullerton, CA
- Architected automated data processing and training pipelines to handle 1,231 3D spatial scans on High-Performance Computing (HPC) clusters. - Utilized SLURM array jobs for parallel scaling across distributed computing nodes, reducing pipeline runtime and experimentation phases by 50% (8h to 4h). - Designed a domain-specific RAG pipeline leveraging token-based chunking, implementing rigorous validation and testing strategies within a secure data framework.	
Machine Learning Engineer (Algorithms & Optimization) <i>Indian Space Research Organisation (ISRO)</i>	Dec 2023 – May 2024 Ahmedabad, India
- Profiled and continuously evaluated computer vision pipelines utilizing C++ and Python, reducing real-world imaging preprocessing latency by 60%. - Implemented and updated machine learning algorithms, engineering a rapid autofocus tool utilizing Laplacian sharpness scoring to achieve fast execution latency. - Collaborated effectively in cross-functional development teams to optimize inference throughput under strict physical compute constraints.	

PROJECTS

LLM Inference Serving System & CI/CD Pipeline <i>vLLM, Docker, Cloud Run, Git</i>	Feb 2025 – Present
- Deployed a 14GB fine-tuned transformer model (DeepSeek) to Google Cloud Run utilizing an Nvidia L4 GPU, optimizing inference for high concurrency and scale. - Bypassed serverless startup limits by engineering CI/CD workflows with custom baked Docker images, reducing cold-boot deployment times to under 30 seconds. - Designed an interactive Streamlit dashboard for system analysis, integrating an autonomous regex-based error parser to continuously monitor and validate compiler logs.	
Predictive ML Dashboard & Cloud Architecture <i>Flask, AWS, React, Python, SQL</i>	Aug 2025
- Developed an end-to-end predictive ML model utilizing 39 engineered features, packaging the solution as a containerized Flask API via Docker. - Designed a React-based dashboard for system analysis and monitoring, allowing cross-functional teams to query operational data in real-time. - Deployed the backend service to AWS with automated load balancing to handle dynamic request spikes, ensuring high reliability for production environments.	

TECHNICAL SKILLS

MLOps & Cloud	Docker, CI/CD Pipelines, Google Cloud Run, AWS, Kubernetes concepts, HPC (SLURM), Git
AI & Inference Serving	vLLM, LLM Inference Systems, DeepSeek, Llama 3.2, PyTorch, RAG Pipelines, Predictive ML
Languages & Development	Python, SQL, C++, Object-Oriented Programming, Agile/Scrum
Dashboards & APIs	Streamlit, React, Flask, REST APIs, System Monitoring, Error Detection